LaTeX Cheat Sheet
Mathematics & Statistics

A comprehensive reference for mathematical and statistical notation in LaTeX

1. Math Modes

Inline math	`$...$`	Example: $x^2 + y^2 = z^2$
Display math	`$$...$$`	Centered equation on new line
Numbered equation	`\begin{equation}...\end{equation}`	Auto-numbered display
Align environment	`\begin{align}...\end{align}`	Multiple aligned equations

2. Greek Letters

Lowercase

α \alpha	β \beta	γ \gamma	δ \delta
ε \epsilon	ζ \zeta	η \eta	θ \theta
ι \iota	κ \kappa	λ \lambda	μ \mu
ν \nu	ξ \xi	ο o	π \pi
ρ \rho	σ \sigma	τ \tau	υ \upsilon
ϕ \phi	χ \chi	ψ \psi	ω \omega

Uppercase

Γ \Gamma	Δ \Delta	Θ \Theta	Λ \Lambda
Ξ \Xi	Π \Pi	Σ \Sigma	Υ \Upsilon
Φ \Phi	Ψ \Psi	Ω \Omega	

3. Basic Operations & Relations

+ Plus	- Minus	× \times	÷ \div
± \pm	∓ \mp	· \cdot	* \ast
= Equals	≠ \neq	≈ \approx	≡ \equiv
< Less	> Greater	≤ \leq	≥ \geq
≪ \ll	≫ \gg	∝ \propto	∞ \infty

4. Fractions, Roots & Powers

Fraction	`\frac{numerator}{denominator}`	x/y → $\frac{x}{y}$
Square root	`\sqrt{x}`	$\sqrt{2}$
nth root	`\sqrt[n]{x}`	$\sqrt[3]{8}$
Superscript	`x^{2}`	x^2, e^{-x}
Subscript	`x_{i}`	x_i, $x_{i,j}$
Both	`x_{i}^{2}`	x_i^2

5. Sums, Products, Limits & Integrals

Sum	`\sum_{i=1}^{n} x_i`	Σ from i=1 to n of x_i
Product	`\prod_{i=1}^{n} x_i`	Π from i=1 to n of x_i
Limit	`\lim_{x \to \infty} f(x)`	Limit as x approaches infinity
Integral	`\int_{a}^{b} f(x) dx`	Definite integral from a to b
Double integral	`\iint_{D} f(x,y) dA`	Double integral over region D
Triple integral	`\iiint_{V} f(x,y,z) dV`	Triple integral over volume V

6. Common Mathematical Functions

sin(x) \sin	cos(x) \cos	tan(x) \tan
log(x) \log	ln(x) \ln	exp(x) \exp
arcsin \arcsin	arccos \arccos	arctan \arctan
sinh \sinh	cosh \cosh	tanh \tanh
max \max	min \min	sup \sup
inf \inf	det \det	dim \dim

7. Brackets & Delimiters

Parentheses	`( )`	Auto-sizing: \left(... \right)
Brackets	`[ ]`	Auto-sizing: \left[... \right]
Braces	`\{ \}`	Auto-sizing: \left\{ ... \right\}
Angle brackets	`\langle \rangle`	■x■
Absolute value	`|x|`	Auto-sizing: \left| x \right|
Floor	`\lfloor x \rfloor`	Floor function

Ceiling	`\lceil x \rceil`	Ceiling function
Norm	`|x|`	Norm: \left| x \right|

8. Matrices & Arrays

Matrix	`\begin{matrix} a & b \\ c & d \end{matrix}`	No brackets
Parentheses matrix	`\begin{pmatrix} a & b \\ c & d \end{pmatrix}`	Round brackets
Brackets matrix	`\begin{bmatrix} a & b \\ c & d \end{bmatrix}`	Square brackets
Braces matrix	`\begin{Bmatrix} a & b \\ c & d \end{Bmatrix}`	Curly braces
Vertical bars	`\begin{vmatrix} a & b \\ c & d \end{vmatrix}`	Determinant
Double bars	`\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}`	Norm notation

Note: Use & to separate columns and \\ to create new rows

9. Statistics Symbols

Mean	`\bar{x} or \mu`	x■ or μ
Variance	`\sigma^2 or s^2`	σ² or s²
Std deviation	`\sigma or s`	σ or s
Expected value	`\mathbb{E}[X] or E[X]`	E[X]
Probability	`\mathbb{P}(A) or P(A)`	P(A)
Covariance	`\text{Cov}(X,Y)`	Cov(X,Y)
Correlation	`\rho or r`	ρ or r
Chi-squared	`\chi^2`	χ²
Binomial coeff	`\binom{n}{k}`	n choose k
Distribution	`X \sim N(\mu, \sigma^2)`	X follows Normal

10. Set Theory

∈ \in	∉ \notin	⊂ \subset	⊆ \subseteq
⊃ \supset	⊇ \supseteq	∪ \cup	∩ \cap
∅ \emptyset	■ \mathbb{N}	■ \mathbb{Z}	■ \mathbb{Q}
■ \mathbb{R}	■ \mathbb{C}	∀ \forall	∃ \exists

11. Logic Symbols

∧ \land (AND)	∨ \lor (OR)	¬ \neg (NOT)	→ \to (implies)
⇒ \Rightarrow	⇔ \Leftrightarrow	↔ \leftrightarrow	∴ \therefore

12. Calculus & Analysis

Partial derivative	`\frac{\partial f}{\partial x}`	∂f/∂x
Total derivative	`\frac{d f}{d x}`	df/dx
Nabla (gradient)	`\nabla f`	∇f
Divergence	`\nabla \cdot F`	∇·F
Curl	`\nabla \times F`	∇×F
Laplacian	`\nabla^2 f or \Delta f`	∇²f or Δf

13. Accents & Font Styles

Hat	`\hat{x}`	x■
Bar	`\bar{x}`	x■
Tilde	`\tilde{x}`	x■
Vector	`\vec{x}`	x■
Dot	`\dot{x}`	■ (time derivative)
Double dot	`\ddot{x}`	■ (2nd derivative)
Bold	`\mathbf{x}`	Bold x
Italic	`\mathit{x}`	Italic x
Calligraphic	`\mathcal{X}`	Script X
Blackboard	`\mathbb{R}`	■ (real numbers)

14. Aligned Equations

Use the `align` environment for multiple equations:

```
\begin{align} y &= mx + b \\ z &= ax^2 + bx + c \\ w &= f(x) \end{align}
```

Use & to align at a specific point (usually =)

15. Piecewise Functions

Use the `cases` environment:

```
f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases}
```

16. Common Statistical Distributions

Normal	`X \sim N(\mu, \sigma^2)`	Normal distribution
Binomial	`X \sim \text{Bin}(n, p)`	Binomial distribution

Poisson	`X \sim \text{Poisson}(\lambda)`	Poisson distribution
Exponential	`X \sim \text{Exp}(\lambda)`	Exponential distribution
Uniform	`X \sim U(a, b)`	Uniform distribution
Chi-square	`X \sim \chi^2(k)`	Chi-squared, k df
t-distribution	`X \sim t(\nu)`	t-distribution, ν df
F-distribution	`X \sim F(d_1, d_2)`	F-distribution

17. Hypothesis Testing

Null hypothesis	`H_0: \mu = \mu_0`	H■
Alternative hypothesis	`H_1: \mu \neq \mu_0`	H■ or H■
Test statistic	`t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}`	t-test
p-value	`p = P(T > t | H_0)`	Probability under H■
Significance level	`\alpha = 0.05`	Type I error rate
Power	`1 - \beta`	Probability of detecting effect

18. Regression

Simple linear	`y = \beta_0 + \beta_1 x + \epsilon`	y = β■ + β■x + ε
Multiple linear	`y = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p + \epsilon`	Multiple predictors
Matrix form	`\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}`	y = Xβ + ε
OLS estimate	`\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}`	β̂ = (X■X)⁻¹X■y
R-squared	`R^2 = 1 - \frac{SS_{res}}{SS_{tot}}`	Coefficient of determination
Adjusted R²	`\bar{R}^2 = 1 - \frac{(1-R^2)(n-1)}{n-p-1}`	Adjusted for predictors

19. Essential LaTeX Packages

`amsmath`	Advanced math environments (align, cases, etc.)
`amssymb`	Additional mathematical symbols
`mathtools`	Extensions to amsmath
`bm`	Bold math symbols (\bm{x})
`siunitx`	Typesetting units and numbers
`booktabs`	Professional tables
`float`	Better figure/table placement

Use in preamble: `\usepackage{package_name}`

20. Tips & Best Practices

• Always use math mode ($...$) for variables, even single letters

• Use \left and \right for automatic delimiter sizing

• Use \text{} for text inside math mode

• Use \, for thin space, \; for medium space in equations

• For multi-line equations, use align instead of multiple equation environments

- Use \cdot for multiplication dot, not period
- Use proper notation: \sin, \cos instead of sin, cos
- For matrices with many elements, consider using \dots, \vdots, \ddots